OT오티

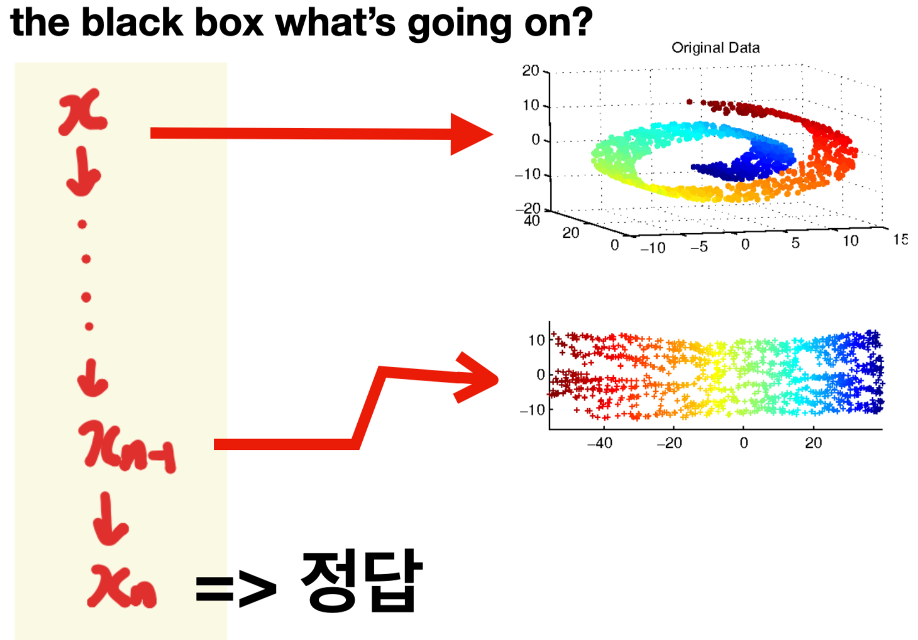
todo: 멋진 것 만들기

수리과학부25 윤승현 + Chatgpt

간단하게 “자기”소개~

RMK: what we have done.

Inside the black box what's going on?



2025 윤승현 1장: Deep Learning Circuit

Q2? on Pointer Network

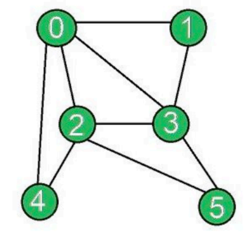
We note that any permutation of the sequence $\mathcal{C}^{\mathcal{P}}$ represents the same triangulation for $\mathcal{P}$, additionally each triangle representation C_i of three integers can also be permuted. Without loss of generality, and similarly to what we did for convex hulls at training time, we order the triangles C_i by their incenter coordinates (lexicographic order) and choose the increasing triangle representation[2]. Without ordering, the models learned were not as good, and finding a better ordering that the Ptr-Net could better exploit is part of future work.

(1, 2, 3) = (1, 3, 2) = (2, 1, 3) …

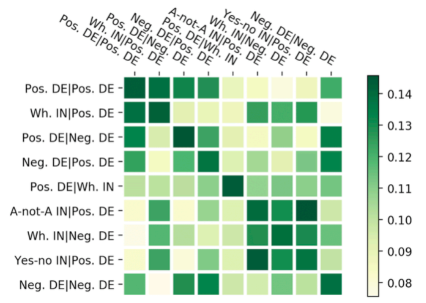
n!의 순열…중에 **무엇**을?

2025 윤승현 RNN: The road to transformer

Prev. Relational Network



	0	1	2	3	4	5
0	0	1	1	1	1	0
1	1	0	0	1	0	0
2	1	0	0	1	1	1
3	1	1	1	0	0	1
4	1	0	1	0	0	0
5	0	0	1	1	0	0



output -> sigma(AHW)

self-attention은 adjacency matrix를 내적 연산으로 만든다!!

2025 윤승현 RNN: The road to transformer

기초적 분석

Ilya list 읽기

직접 질문 던지기

RMK: what we have done.

- [x] Convolutional Neural Networks(5/5) ✅
- [x] Recurrent Neural Networks(10/10) ✅
- [ ] Transformers(0/3)
- [ ] Information Theory(0/5)
- [ ] Miscellaneous(0/4)

주의: 추후 진행될 시그를 위해 필요한 것

1순위: 열ㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹㄹ정 (오늘 오심!)

2순위: 수학으로 딥러닝의 기초 이해… (오늘 할거임!)

n-1순위: 뛰어난 수학 실력

n순위: 코딩 실력

(있으면 좋고,,,)

왜 수학인가..에 대한 초보의 대답

정보란 무엇이지? - logp(x)

1	0	1	1

1. bit를 생각해보자!

2. 정보=bit?

3. 메세지를 보내야한다..(엔트로피)

4. 스무고개, wordle?

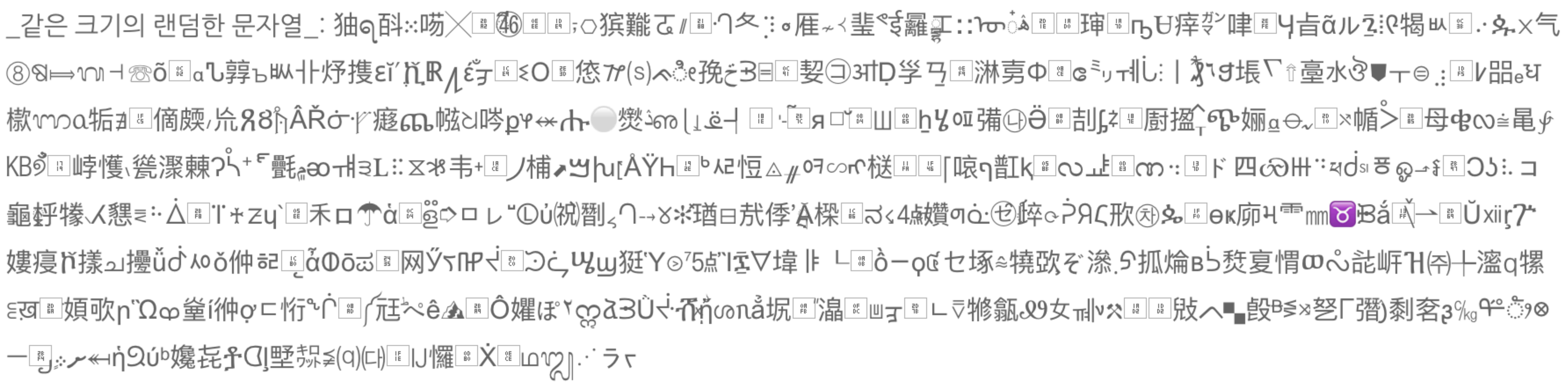
5. 몬티홀 문제를 엔트로피로 생각해보자. 1.58bit -> 0.91bit

왜 수학인가..에 대한 초보의 대답

hello worl_ olleh lrow_

모델링이란 무엇인가?

의미있는 패턴의 발견(=압축)



	원본	압축후
지금까지 글	1620	1003
같은 크기의 랜덤한 문자열	1620	1388

왜 수학인가..에 대한 초보의 대답

hello worl_ olleh lrow_

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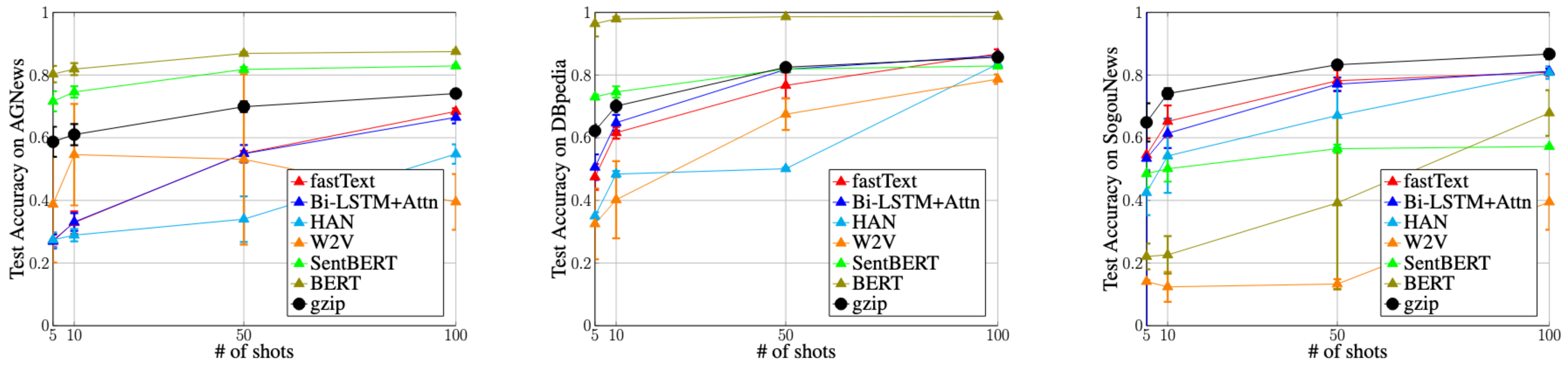


Figure 2: Comparison among different methods using different shots with 95% confidence interval over five trials.

gzip + knn > bert?

“Low-Resource” Text Classification: A Parameter-Free Classification Method with Compressors(2023 ACL)

왜 수학인가..에 대한 초보의 대답

hello worl_ olleh lrow_

모델링이란 무엇인가?

의미있는 패턴의 발견(=압축)

정보란 무엇인가?

정보를 어떻게 수학적으로 표현?
정보를 적은 비용으로 인코딩?

Since then a new method has been found for estimating these quantities, which is more sensitive and takes account of long range statistics, influences extending over phrases, sentences, etc. This method is based on a study of the predictability of English; how well can the next letter of a text be predicted when the preceding N letters are known. The results of some experiments in prediction will be given, and a theoretical analysis of some of the properties of ideal prediction. By combining the experimental and theoretical results it is possible to estimate upper and lower bounds for the entropy and redundancy. From this analysis it appears that, in ordinary literary English, the long range statistical effects (up to 100 letters) reduce the entropy to something of the order of one bit per letter, with a corresponding redundancy of roughly 75%. The redundancy may be still higher when structure extending over paragraphs, chapters, etc. is included. However, as the lengths involved are increased, the parameters in question become more

[1] C. E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal*, v. 27, pp. 379–423, 623–656, July, October, 1948.

50

Shannon, Prediction and Entropy of Printed English (1950)

왜 수학인가..에 대한 초보의 대답

(1) THE ROOM WAS NOT VERY LIGHT A SMALL OBLONG
(2) ----ROO------NOT-V-----I------SM----OBL----
(1) READING LAMP ON THE DESK SHED GLOW ON
(2) REA----------O------D----SHED-GLO--O--
(1) POLISHED WOOD BUT LESS ON THE SHABBY RED CARPET
(2) P-L-S-----O---BU--L-S--O------SH-----RE--C------

reduced text

TABLE I

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	100
1	18.2	29.2	36	47	51	58	48	66	66	67	62	58	66	72	60	80
2	10.7	14.8	20	18	13	19	17	15	13	10	9	14	9	6	18	7
3	8.6	10.0	12	14	8	5	3	5	9	4	7	7	4	9	5	
4	6.7	8.6	7	3	4	1	4	4	4	4	5	6	4	3	5	3
5	6.5	7.1	1	1	3	4	3	6	1	6	5	2	3			4
6	5.8	5.5	4	5	2	3	2			1	4	2	3	4	1	2
7	5.6	4.5	3	3	2	2	8		1	1	1	4	1		4	1
8	5.2	3.6	2	2	1	1	2	1	1	1	1		2	1	3	
9	5.0	3.0	4		5	1	4		2	1	1	2		1		1
10	4.3	2.6	2	1	3		3	1					2			
11	3.1	2.2	2	2	2	1			1	3		1	1	2	1	
12	2.8	1.9	4		2	1	1	1			2	1	1		1	1
13	2.4	1.5	1	1	1	1	1	1	1	1		1	1			
14	2.3	1.2		1			1					1				1
15	2.1	1.0	1	1							1	1	1			
16	2.0	.9					1			1					1	
17	1.6	.7	1		2	1	1				1		2	2		
18	1.6	.5													1	

Shannon, Prediction and Entropy of Printed English (1950)

왜 수학인가..에 대한 초보의 대답

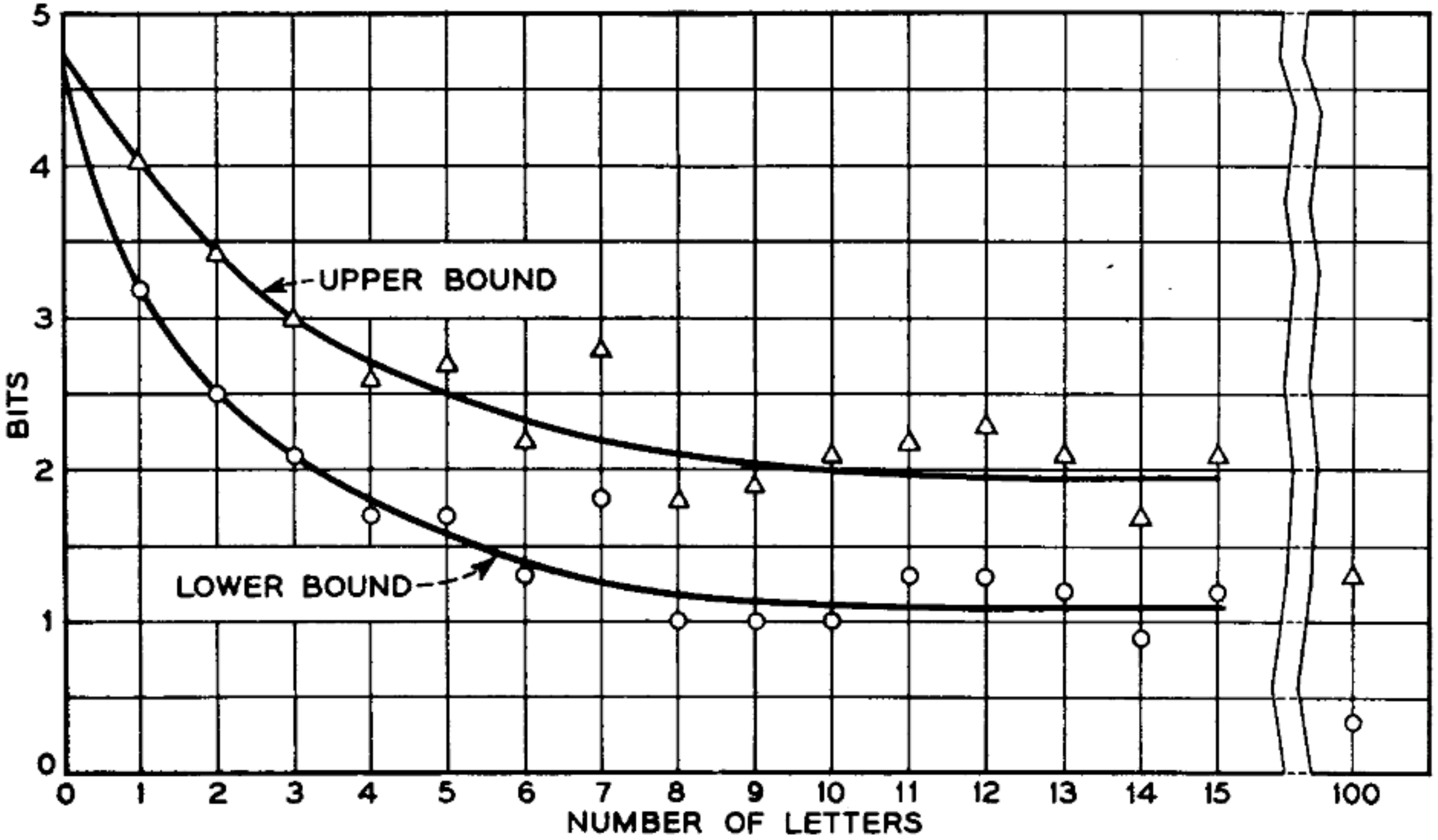
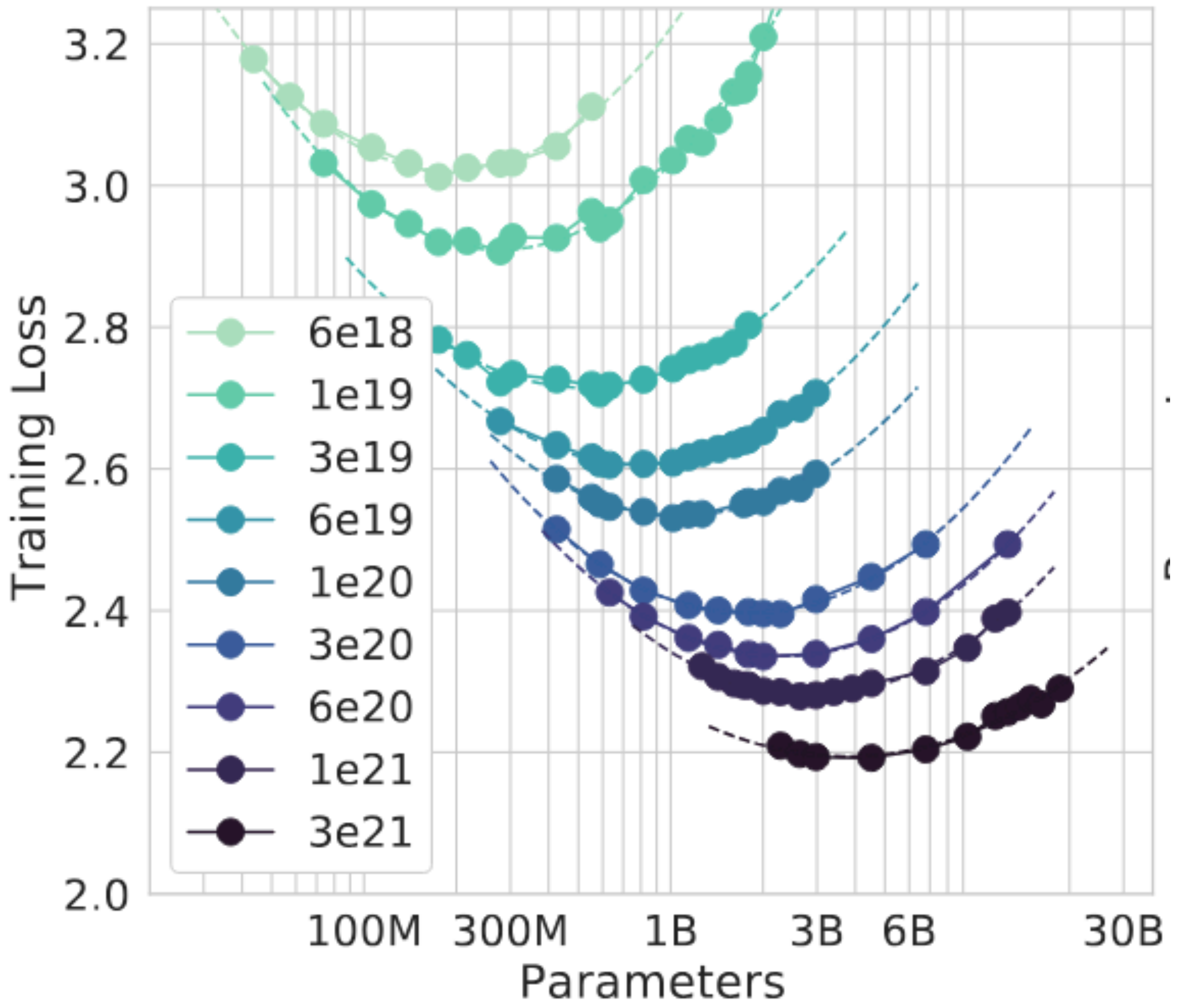


Fig. 4—Upper and lower experimental bounds for the entropy of 27-letter English.

내가 뭐라고 했을까요

Shannon, Prediction and Entropy of Printed English (1950)

왜 수학인가..에 대한 초보의 대답



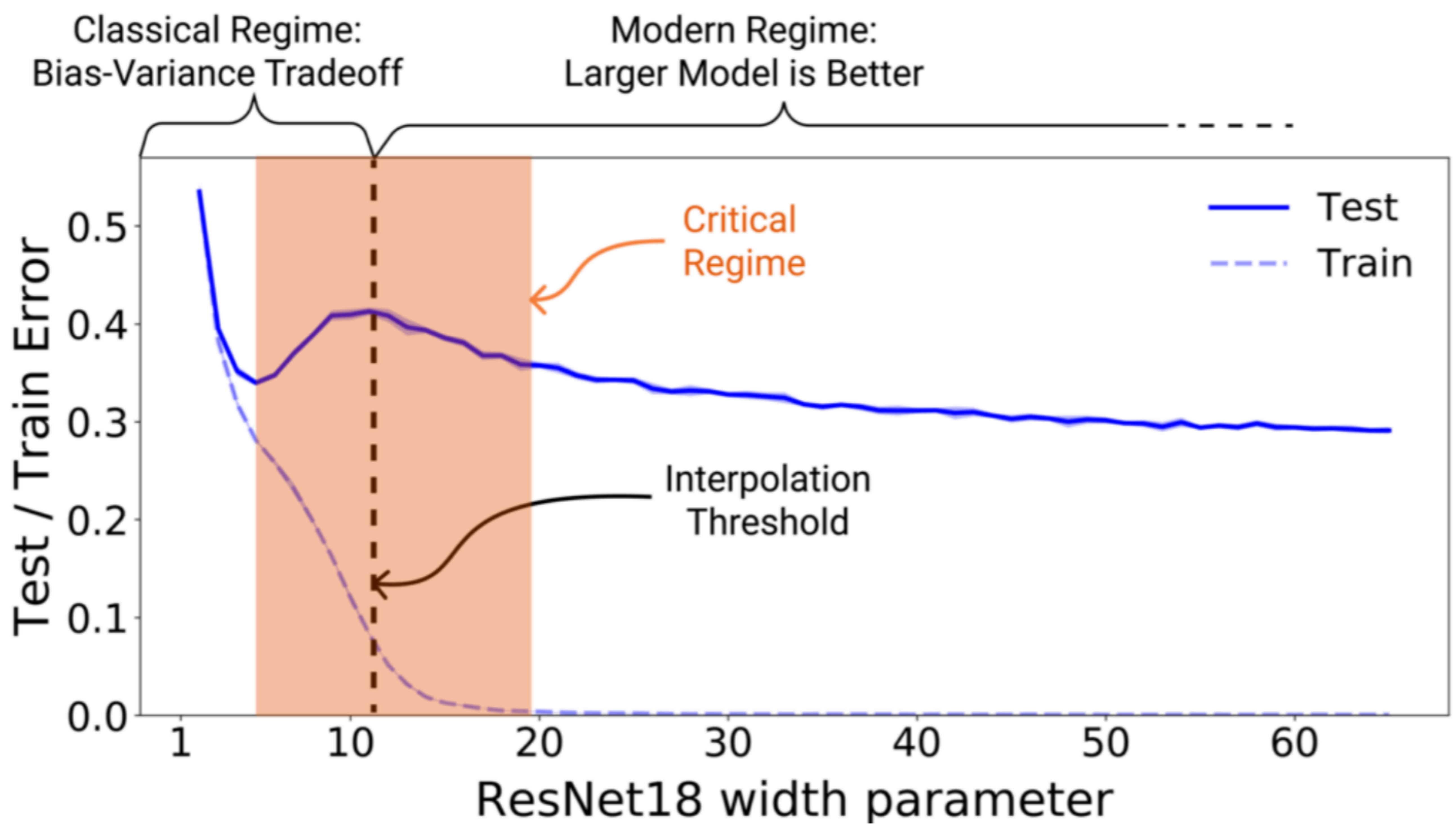
$$L(D, N) = 2.62 + \frac{A}{N^a} + \frac{B}{D^b}$$

더 이상 나아질 수 없는 임계점 entropy 2.62bit

좀 전의 실험결과와는 조금 다르네요

Training Compute-Optimal Large Language Models (2022, Deepmind) / Chinchilla Scaling: A replication attempt(2024, Epoch AI)

왜 수학인가..에 대한 초보의 대답



overfitting(옛말)

Grokking/Generalization

Deep Double Descent: Where bigger model and more data hurt (OpenAI)

왜 수학인가..에 대한 초보의 대답

정보란 무엇이지? - logp(x)

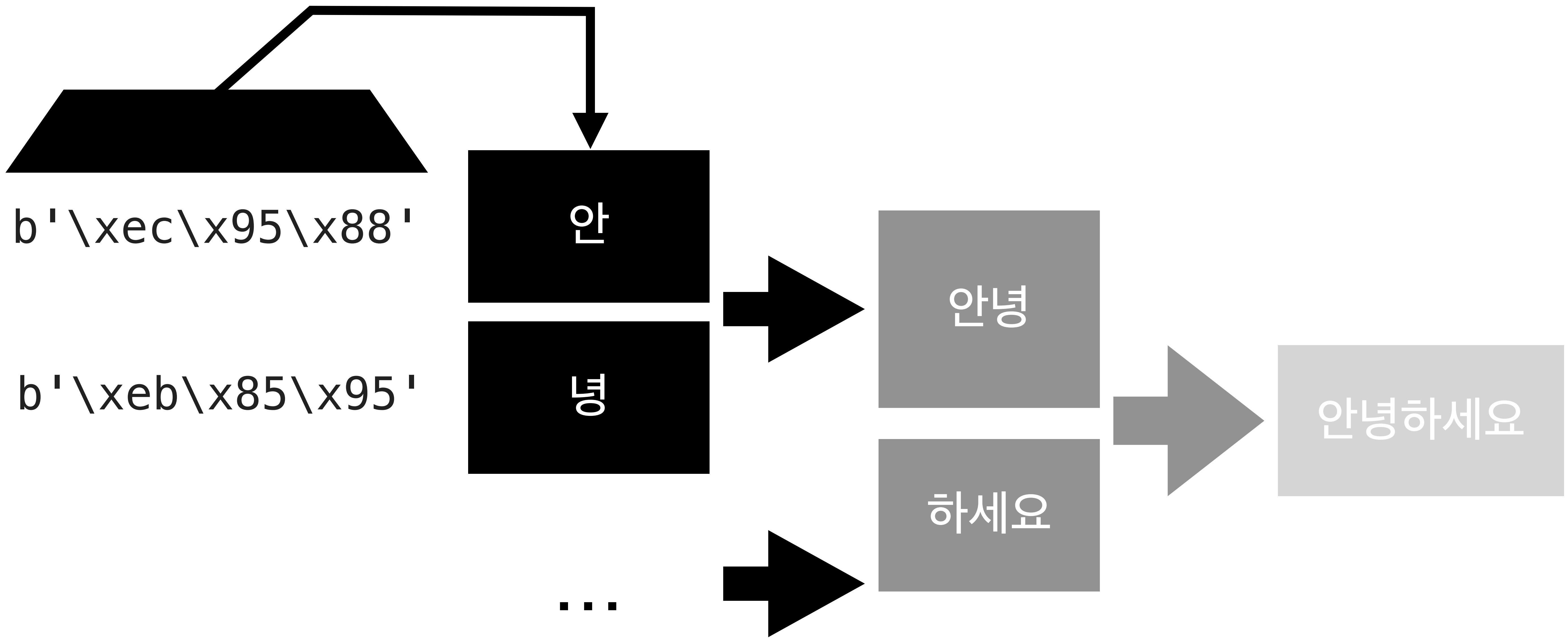
다른 distribution 간의 거리는 어떻게? - KL divergence/CE

이산적 기호 집합 말고 associative한 파라미터의 정보 저장 용량은?

정보 이론(Information Theory)

double descent, grokking, hopfield net, data, tokenizing…

문자란 무엇일까? for representing as vector

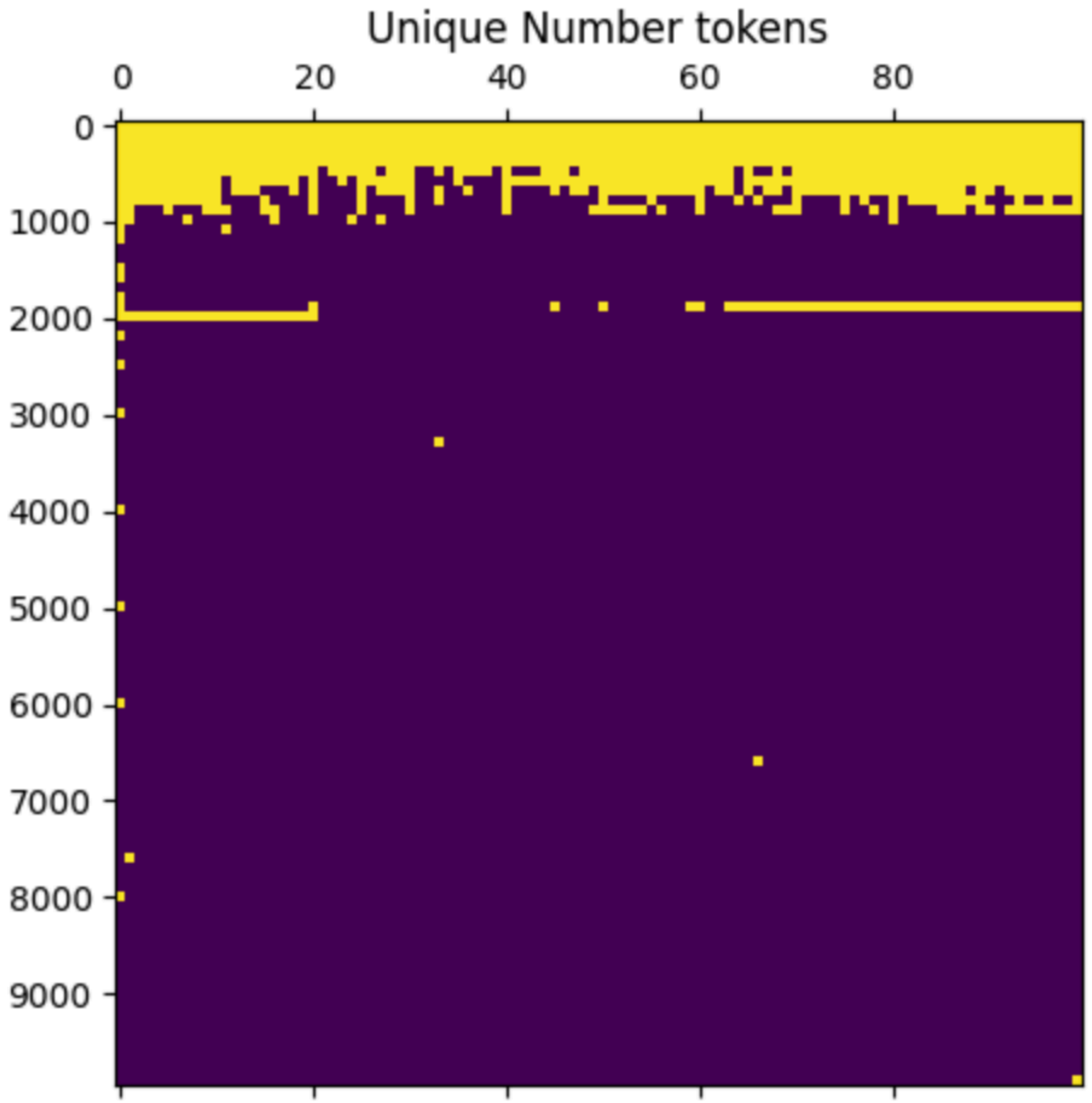


Low-entropy 쌍을 융합

= 응집성 높은 친구들의 합성

= 문자 ~~ 압축

왜 우리는 생각해야 하는가? - 숫자 표현 예시(소수는?)



Plot of unique tokens in the first 10k integers for the GPT2 tokenizer.

199x는 역사에 기록된 것 많음!

-> unique token

(1) + (199x)는 우리의 생각처럼 작동하지 않는다

덧셈 과정의 근본에 대한 학습보다는 그저 sequence 암기가 될 가능성이 큼

coherent number representation

Integer tokenization is insane(2023, Beren)